

Independent Audit Note on “Cobham-Free Full General Automatic Rigidity for Collatz-Invariant Sequences”

Adversarial examination of cross-base kernel transfer,
dyadic thinness, and topological pumping

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24 September 2026

Revised post-release supplementary audit note

Abstract

This note records a revised post-release adversarial audit of four structurally important points in the Cobham-free Full General ARC proof. The audit targets (A1) transfer from base-2 finite-kernel automaticity to base-3 automaticity of the odd subsequence, (A2) dyadic nowhere-thickness of the odd-subsequence change set, (A3) the odd-base topological-pumping step from dyadic nowhere-thickness to finite support, and (A4) the possibility that the main rigidity theorem is vacuous or that the exclusion of zero is not genuinely sharp.

All four audit items have now been reconstructed at the paper level and encoded in separate Lean 4 audit files under an explicit independence scope. A1 avoids the earlier packaged transfer result; A2 reuses the lower local-density theorem but independently rebuilds the A2-specific affine pullback; A3 directly avoids the packaged Stage 6–10 conclusions while reusing lower infrastructure. Lean kernel type-checking completed successfully for every audit file, and each audit module also completed its targeted `lakebuild`. For the principal audit endpoints, `\#printaxioms` reported only the standard Lean/Mathlib dependencies `propext`, `Classical.choice`, and `Quot.sound`, with no `sorryAx`. No counterexample or confirmed defect in the published Full General ARC theorem was found.

The strongest result of the revised audit is therefore not merely a source trace. A1 is re-derived from the kernel definitions using the primitive-root behavior of 2 modulo powers of 3; A2 is rebuilt from the lower local-density theorem by an explicit affine pullback $x = 2n + 1$; A3 reconstructs the odd-base pumping-to-dyadic-filling contradiction; and A4 gives an explicit finite-kernel shortcut-invariant witness in every base $B \geq 2$ whose value at zero may differ from its positive value.

Here “independent” means independent of the corresponding packaged proof endpoint and reconstructed through lower-level ingredients. It does not mean external third-party peer review, nor does it claim complete import-graph isolation of every lower lemma reused by the audit files.

1 Status and purpose

The parent paper proves the following rigidity statement for the shortcut Collatz map

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n + 1)/2, & n \equiv 1 \pmod{2}. \end{cases}$$

For every base $B \geq 2$, if $a : \mathbb{N}_0 \rightarrow \alpha$ has finite base- B kernel and satisfies

$$a(T(n)) = a(n) \quad (n \in \mathbb{N}_0),$$

then a is constant on the positive integers. The value at zero may remain independent.

The first version of this note identified four audit targets. The revised audit has now carried each one through an independent mathematical reconstruction and a separate Lean verification. In this note the word *closed* means that the specific audit objection has been answered by a reconstruction that was checked by Lean; it does not mean that no further mathematical or literature review could ever be useful.

Audit target	Revised finding	Status
A1. Odd subsequence becomes base-3 finite-kernel automatic	A fresh kernel-level proof embeds every base-3 residual of $b(n) = a(2n + 1)$ into the finite base-2 kernel of a . The reconstruction does not invoke <code>arc_three_kernel_odd_finite</code> , <code>arc_three_kernel_odd_subset</code> , or <code>arc_lemma3_1</code> .	CLOSED: independently reconstructed and Lean-verified.
A2. Change set is dyadically nowhere thick	The affine pullback $x = 2n + 1$ is rebuilt directly from the lower local-density theorem with gap $H = 2$, without calling the packaged A2 child or thinness theorems.	CLOSED: independently reconstructed within the stated audit scope and Lean-verified.
A3. Odd-base automatic + dyadically nowhere thick implies finite support	Residue permutation, dyadic child filling, collision with nowhere-thickness, and the finite-support conclusion were reconstructed in a separate audit chain without calling the packaged Stage 6–10 conclusions under audit.	CLOSED: direct non-use reconstruction within the stated audit scope and Lean-verified.
A4. Vacuous truth / sharpness at zero	For every $B \geq 2$, the explicit sequence $a(0) = x$, $a(n) = y$ for $n > 0$ has a kernel of at most two states and is shortcut invariant. Choosing $x \neq y$ proves genuine freedom at zero.	CLOSED: constructively and Lean-verified.

2 Audit A1: independent base-3 kernel transfer

Let $a : \mathbb{N}_0 \rightarrow \alpha$ satisfy the two shortcut branch identities

$$a(2n) = a(n), \quad a(2n + 1) = a(3n + 2),$$

and define

$$b(n) = a(2n + 1).$$

The target is

$$\mathcal{K}_2(a) \text{ finite} \implies \mathcal{K}_3(b) \text{ finite}.$$

The revised audit re-derives this implication without calling the earlier packaged ARC2 transfer theorem.

2.1 Independent paper reconstruction

Take an arbitrary base-3 kernel residual of b ,

$$v(n) = b(3^e n + r), \quad 0 \leq r < 3^e.$$

Using the odd branch once,

$$\begin{aligned} v(n) &= a(2(3^e n + r) + 1) \\ &= a(3^{e+1} n + (3r + 2)). \end{aligned}$$

Set $j = e + 1$ and $c = 3r + 2$. Then

$$0 < c < 3^j, \quad 3 \nmid c.$$

Because 2 is a primitive root modulo 3^j , choose $s \geq 0$ such that

$$2^s c \equiv -1 \pmod{3^j}.$$

Hence

$$2^s c + 1 = q 3^j$$

for some integer $q \geq 1$. Repeated use of the even identity together with the odd identity in reverse gives

$$a(3^j n + c) = a(2^{s+j} n + R), \quad R = 2^j q - 1.$$

The same arithmetic gives

$$0 \leq R < 2^{s+j}.$$

Therefore every base-3 residual v of b is literally a base-2 kernel residual of a . Thus

$$\mathcal{K}_3(b) \subseteq \mathcal{K}_2(a),$$

and the finiteness of $\mathcal{K}_2(a)$ implies the finiteness of $\mathcal{K}_3(b)$. No Cobham theorem, Collatz convergence statement, or finite-alphabet hypothesis is used in this reconstruction.

2.2 Independent Lean audit

The separate file

`ARCA1IndependentAudit_v2.lean`

formalizes this reconstruction and deliberately avoids direct invocation of `arc_three_kernel_odd_finite`, `arc_three_kernel_odd_subset`, and `arc_lemma3_1`. The standalone command

`lakeenvleanARC/ARCA1IndependentAudit_v2.lean`

completed successfully, followed by a successful targeted module build (8767 jobs in the recorded project state). The audited endpoints report only `propext`, `Classical.choice`, and `Quot.sound`.

2.3 Audit conclusion for A1

The original source-level transfer has now been backed by a second proof route at both the paper and Lean levels. The specific concern that the base-3 odd-subsequence automaticity was being supplied by an opaque or convergence-like assumption is therefore closed for the audited formal statement.

3 Audit A2: independent dyadic thinness reconstruction

Define the odd subsequence and its adjacent-change set by

$$b(n) = a(2n + 1), \quad \mathcal{D}_b = \{n \in \mathbb{N}_0 : b(n) \neq b(n + 1)\}.$$

The concern is the interaction between shortcut-Collatz arithmetic and dyadic residue classes. The revised audit isolates exactly the A2-specific affine step and rebuilds it from the lower local-density theorem.

3.1 Independent affine pullback

Start with an arbitrary dyadic parent class

$$n \equiv r \pmod{2^m}.$$

The affine map

$$x = 2n + 1$$

sends this parent into

$$x \equiv 2r + 1 \pmod{2^{m+1}}.$$

Apply the already established base-independent local-density theorem to this odd affine image with gap $H = 2$. It supplies a deeper integer dyadic child on which

$$a(x) = a(x + 2).$$

The audit then constructs explicitly a deeper child in the original n -variable whose entire affine image lies in that local-density child. On that child,

$$a(2n + 1) = a(2n + 3),$$

so

$$b(n) = b(n + 1).$$

After converting the integer cylinder to a natural residue class, the child is nonempty, remains inside the prescribed parent, and is disjoint from $\mathcal{D}_b$. This is exactly dyadic nowhere-thickness.

3.2 Independent Lean audit

The file

`ARCA2IndependentAudit.lean`

reconstructs the A2-specific argument without calling `arc2OddSubseq_local_gap_two_child`, `arc2OddSubseq_changeSet_nowhereThick`, or `arc2CobhamRemoval_stage1_thinness`. It reuses the lower theorem `arc5_naturalInvariantBlock_in_dyadicCylinder`, which belongs to the independently developed local-density layer rather than to the A2 conclusion under audit.

The standalone Lean check completed successfully, and the targeted build `lakebuildARC.ARCA2IndependentAudit` completed successfully with 8859 jobs in the recorded project state. The endpoints `arcA2_affine_gap_two_child_independent`, `arcA2_changeSet_nowhereThick_independent`, and `arcA2_target` report only the standard dependencies `propext`, `Classical.choice`, and `Quot.sound`.

3.3 Audit conclusion for A2

The affine conversion is no longer supported merely by source tracing: it has been reconstructed as a separate formal argument from the lower local-density interface. The specific A2 objection is therefore closed. A deeper audit of the local-density theorem itself would be a different audit target, not a gap in the A2 pullback.

4 Audit A3: independent odd-base topological-pumping reconstruction

Let $q \geq 3$ be odd and let $s : \mathbb{N}_0 \rightarrow \{0, 1\}$ have finite base- q kernel. The target statement is that dyadic nowhere-thickness of the support forces the support to be finite. This is the most structurally delicate mixed-base step: the automaton is organized by powers of q , while thinness is organized by powers of 2.

4.1 Independent paper reconstruction

Assume the support is infinite. A finite MSD automaton then has a positive canonical support word longer than its number of states. Pumping yields a nonempty loop block. Doubling that loop gives a pumped numerical family

$$N_k = N_0 + dG_k, \quad G_k = 1 + Q + \cdots + Q^{k-1}, \quad Q = q^{2L},$$

where $L > 0$ is the loop length. Positivity gives $d > 0$, so write

$$d = 2^s(2v + 1).$$

Since q is odd,

$$Q = q^{2L} \equiv 1 \pmod{4}.$$

If

$$j - i = 2^t(2u + 1),$$

the doubled-loop geometric-sum arithmetic gives an exact dyadic displacement

$$N_j - N_i = 2^{s+t}(2z + 1).$$

Consequently, for $0 \leq i < j < 2^M$, the normalized pump residues are distinct modulo 2^M : the valuation t of $j - i$ is always $< M$. There are exactly 2^M indices and 2^M residue classes, so the normalized pump map is a permutation of all residues modulo 2^M .

Restoring the factor 2^s and the offset N_0 , the pumped support family hits every deeper dyadic child of one parent cylinder. Dyadic nowhere-thickness requires an empty positive deeper child inside every parent cylinder. The two statements contradict each other. Hence an infinite odd-base automatic support cannot be dyadically nowhere thick, and the support must be finite.

4.2 Independent Lean audit

The file

ARCA3IndependentAudit.lean

reconstructs the sensitive residue-permutation, child-filling, collision, and finite-support chain with new audit theorem names. It deliberately does not call the packaged Stage 6 residue-permutation theorem, Stage 7 child-filling theorem, Stage 8 collision theorem, Stage 9 topological-pumping endpoint, or Stage 10 kernel endpoint.

The audit does reuse lower ingredients, notably the finite-kernel-to-MSD bridge, the exact Stage-5 dyadic pumping handoff, and a generic long-canonical-word lemma. Because the imported file containing the generic long-word lemma also transitively imports later stages, this audit should be described precisely as *direct non-use reconstruction* of the Stage 6–10 conclusions rather than complete dependency-graph isolation.

The standalone Lean check completed successfully. The targeted command `lakebuildARC.ARC.A3IndependentAudit` then completed successfully with 8819 jobs in the recorded project state. The principal audit endpoints, including `arcA3_oddPumpResidueMap_injective`, `arcA3_evenLoopPumpedFamily_hits_every_child_of_exact_two_factor`, `arcA3_child_filling_not_nowhereThick`, `arcA3_kernelAutomatic_nowhereThick_support_finite`, and `arcA3_target`, report only `propext`, `Classical.choice`, and `Quot.sound`.

4.3 Auxiliary Cantor-style sanity check

The earlier adversarial candidate

$$C = \left\{ \sum_{i \geq 0} \varepsilon_i 3^i : \varepsilon_i \in \{0, 1\}, \text{ finitely many } \varepsilon_i \neq 0 \right\}$$

was also tested by exact finite residue enumeration. For

$$C_L = \left\{ \sum_{i=0}^{L-1} \varepsilon_i 3^i : \varepsilon_i \in \{0, 1\} \right\},$$

finite L already covered every residue modulo 2^m throughout the tested range. This computation is not part of the proof, but it points in the same direction as the formal child-filling mechanism and did not produce a counterexample.

4.4 Audit conclusion for A3

The mixed-base step has now been reconstructed independently at the mathematical level and through a separate Lean chain. No counterexample was found, and the specific objection that the proof merely assumes compatibility between odd-base automaticity and dyadic topology is closed for the audited theorem.

5 Audit A4: constructive non-vacuity and sharpness at zero

The final audit item asks whether the theorem could be true only because its hypotheses describe an empty class, and whether excluding zero from the rigidity conclusion is genuinely necessary.

For arbitrary values $x, y \in \alpha$, define

$$a(0) = x, \quad a(n) = y \quad (n > 0).$$

For every base $B \geq 2$, every base- B kernel residual of this sequence is either the original sequence or the constant sequence $n \mapsto y$. Hence

$$|\mathcal{K}_B(a)| \leq 2.$$

Also $T(0) = 0$ and $T(n) > 0$ for every $n > 0$, so

$$a(T(n)) = a(n) \quad (n \geq 0).$$

If $x \neq y$, then $a(0) \neq a(1)$, while all positive values are equal. Thus the hypothesis class is nonempty in every base and the freedom at zero is real.

5.1 Independent Lean audit

The file

`ARCA4IndependentAudit_v2.lean`

proves the kernel bound and shortcut invariance directly for arbitrary $B \geq 2$, without invoking an earlier ARC sharpness theorem or the Full General rigidity endpoint. It also packages a concrete Boolean witness in `arcA4_bool_witness`.

The standalone Lean check completed successfully; after one local proof repair in the first draft, the final file contains no unsolved goal and no `sorryAx`. The targeted build `lakebuildARC.ARCA4IndependentAudit_v2` completed successfully with 8845 jobs in the recorded project state. The principal endpoints report only the standard Lean/Mathlib dependencies.

5.2 Audit conclusion for A4

The vacuity concern and the sharpness-at-zero concern are both closed constructively. This is stronger than merely exhibiting one base-specific example: the same two-state kernel construction works uniformly for every base $B \geq 2$.

6 What a genuine counterexample would have to provide

To refute the Full General ARC theorem, it is not enough to propose a coloring that is constant on Collatz orbits, or a sparse automatic set, or a coloring that respects only the even branch. A genuine counterexample must explicitly exhibit some $B \geq 2$ and a sequence $a : \mathbb{N}_0 \rightarrow \alpha$ satisfying all three conditions simultaneously:

- (i) the full base- B kernel $\mathcal{K}_B(a)$ is finite;
- (ii) $a(T(n)) = a(n)$ for every $n \geq 0$;
- (iii) there exist positive m, n with $a(m) \neq a(n)$.

For example, assigning colors to hypothetical nonstandard Collatz components might satisfy (ii), but it says nothing by itself about the finite-kernel condition (i). Likewise, extending a coloring by $a(2^k q) = a(q)$ enforces the even relation but does not automatically enforce the odd relation

$$a(2n + 1) = a(3n + 2).$$

Any claimed counterexample must survive these exact checks.

7 Lean 4 verification summary and audit scope

The revised audit adds four dedicated source files to the already completed Cobham-free Full General ARC development:

Audit file	Main independent target	Target build
ARCA1IndependentAudit_v2.lean	base-2 kernel $\Rightarrow$ base-3 odd-subsequence kernel transfer	8767 jobs
ARCA2IndependentAudit.lean	affine pullback $x = 2n + 1$ and dyadic thinness of the odd change set	8859 jobs
ARCA3IndependentAudit.lean	odd-base residue permutation $\Rightarrow$ child filling $\Rightarrow$ finite support	8819 jobs
ARCA4IndependentAudit_v2.lean	all-base finite-kernel invariant witness and sharpness at zero	8845 jobs

Each file first passed `lakeenvlean` as a standalone module and then its corresponding targeted `lakebuild`. Across the principal audit endpoints, `\#printaxioms` reports the standard dependencies

`propext, Classical.choice, Quot.sound`

with the minor exception that some elementary endpoints require only a subset of these. No final audit endpoint reports `sorryAx`.

This evidence has a precise scope. Lean verifies the formal statements from the definitions and imported lower lemmas. The audit files were designed to avoid the packaged conclusions being audited, but they intentionally reuse lower infrastructure rather than re-proving all of Mathlib, the automaticity bridge, local-density arithmetic, or Stage-5 pumping from first principles. Accordingly, this revised note supports a strong internal consistency claim for A1–A4, not a claim of external peer review, complete foundational independence, or exhaustive novelty.

8 Current audit judgment

The four objections isolated in the first version of this note have now all been carried through a second mathematical reconstruction route, with the scope of reused lower lemmas stated explicitly, and separate Lean kernel verification:

1. **A1 CLOSED.** The base-3 odd-subsequence kernel transfer was independently re-derived from the kernel definitions and formalized without the earlier packaged transfer lemma.
2. **A2 CLOSED.** The affine dyadic pullback was independently rebuilt from the lower local-density theorem and Lean-verified.
3. **A3 CLOSED.** The odd-base residue-permutation and dyadic child-filling contradiction was independently reconstructed and Lean-verified through a direct non-use audit chain.
4. **A4 CLOSED.** Non-vacuity and the genuine freedom of the zero value were established uniformly for every base $B \geq 2$ by an explicit two-state-kernel witness.

No confirmed mathematical defect, hidden Collatz-convergence premise, Cobham handoff, or `sorryAx` was found in these audited steps. This conclusion should still be read with ordinary mathematical caution: future review may probe lower lemmas, modeling choices, or literature overlap that are outside the four objections closed here.

Accordingly, this document remains an *Independent Audit Note*, not an erratum or corrigendum. Its revised purpose is to record that the four initially identified attack points have survived a second reconstruction route. If a genuine defect is later identified, it should be documented separately and transparently as a correction.

Revision record

First audit note: 23 September 2026. Revised audit: 24 September 2026 after completion of the independent A1–A4 paper reconstructions and Lean audit modules. The original parent theorem is unchanged; this revision updates the status of the supplementary audit only.

Reproducibility note

The finite Cantor-style residue experiment used the exact sets

$$C_L \bmod 2^m$$

formed by all 2^L choices of digits $\varepsilon_i \in \{0, 1\}$, with set compression after each added power of 3. The reported smallest L values are therefore exhaustive for the stated moduli. They are included only as an adversarial sanity check and are not used in the Lean proof.

Acknowledgment of AI assistance

Generative AI systems were used to formulate adversarial objections, inspect proof architecture, assist with Lean-source tracing, run finite computational checks, and prepare this note. An AI-generated objection is not treated as a mathematical result unless its claim is independently verified. Responsibility for the released mathematical statements and the interpretation of the formal artifacts remains with the human author.

References

- [1] Muteki, *Cobham-Free Full General Automatic Rigidity for Collatz-Invariant Sequences*, Full General ARC release v1.0.0, 24 September 2026.
- [2] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [3] A. Cobham, On the base-dependence of sets of numbers recognizable by finite automata, *Mathematical Systems Theory* **3** (1969), 186–192.